Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	AI-Powered Cyber-Physical Integrated Training Platform for Emergency Vascular and Trauma Care			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio・Medical		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Information & System	Hospital Information Systems and Equipment	Human Resource Competency Development Services	Intelligent Learning Support / Management Technologies
Concept of Proposed Technology	**Core Concept** The proposed technology is an **AI-powered cyber-physical integrated emergency training platform** designed to enable paramedics and novice medical personnel to perform **high-difficulty, life-saving emergency procedures** in disaster and extreme environments. The platform combines **high-fidelity bio-mimetic physical simulators**, equivalent to human anatomical structures, with **real-time AI-based guidance and evaluation**, enabling non-expert responders to rapidly and accurately perform critical trauma interventions during the "golden hour." The platform is specifically designed for **massive hemorrhage control, emergency vascular stabilization, suturing, intravenous fluid and blood administration, and infection control**, all within a unified, immersive training environment.			

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	Specific Scope and Technical Focus **1. Sutureless Rapid-Connect Training System for High-Risk Trauma** • Development of a sutureless vascular connection training module based on **multi-branched bio-mimetic scaffolds**with a **support-free internal structure** • Establishment of a high-precision training environment using **high-friction interface mechanisms**, enabling non-expert trainees to restore simulated blood flow within **15 minutes** **2. Immersive Human Simulation and Integrated Care Modules** • Integration of emergency response scenarios including vascular stabilization, suturing, intravenous fluid and blood administration, and infection control • Use of **high-precision simulators** replicating cadaver-level anatomical structure and tissue resistance **3. AI-Driven Simulation Guidance and Analytical Engine** • Development of AI algorithms that track user actions in real time, analyze procedural accuracy, and provide immediate corrective feedback • Construction of standardized and personalized training analytics aligned with internationally recognized disaster and trauma care guidelines **Expected Applications** • Disaster response and mass-casualty training for paramedics and first responders • Emergency trauma training for novice medical personnel • Standardized training programs for public healthcare and disaster preparedness systems

Rationale for Joint R&D Project	Massive hemorrhage remains the leading cause of preventable death in large-scale disasters. Existing emergency training methods rely primarily on simplified models that do not adequately reproduce the mechanical complexity, time pressure, or procedural difficulty of real disaster environments.
	
	This project requires **international joint R&D** to overcome technical thresholds in three key areas:
	
	1. **Disaster-Optimized Integrated Training Systems**
	  ○ Integration of high-fidelity bio-mimetic manufacturing with advanced AI-based procedural evaluation
	  ○ Validation of hydrodynamic stability and mechanical reliability in multi-branched, support-free vascular simulation structures
	2. **Global Standardization and Technology Validation**
	  ○ Optimization of sutureless high-friction connection mechanisms according to international medical and trauma care standards
	  ○ Validation of AI-based feedback systems using global-level trauma datasets
	3. **Establishing Market Entry Barriers**
	  ○ Creation of a de facto international training standard for disaster medical education
	  ○ Preparation for global public procurement and disaster relief markets through joint validation with overseas medical institutions

Key Words	• AI-Based Disaster Trauma Training Platform • Support-Free Multi-Branch Grafting • High-Fidelity Bio-Mimetic Simulation • Sutureless Rapid-Connect Mechanism • Golden- Hour Hemorrhage Control Standard

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (6)	End of the Project	Level (8)

R&D Objectives	The objective of this project is to develop an **AI-integrated emergency trauma training platform** and **high-precision bio-mimetic vascular connection technology** optimized for disaster response.
	
	Key development objectives include:
	
	1. **AI-Based Emergency Decision and Real-Time Procedural Analysis**
	  ○ Development of AI algorithms that assist trainees in prioritizing emergency interventions and evaluating procedural accuracy in real time
	  ○ Validation using global disaster and trauma medical datasets
	2. **High-Precision Bio-Mimetic Sutureless Connection Technology**
	  ○ Development of a multi-branch vascular simulation module enabling non-expert trainees to complete vascular stabilization within **15 minutes**
	  ○ Demonstration of mechanical stability without internal structural supports
	3. **Integrated Digital-Physical Simulation Platform**
	  ○ Development of an integrated hardware-software system providing realistic tactile feedback for suturing, intravenous access, and blood transfusion
	  ○ Integration with cadaver-level bio-mimetic simulators			

R&D Target Specifications	• Emergency vascular reconstruction time: **≤ 15 minutes** • Burst pressure resistance of sutureless connection: **≥ 300 mmHg** • Bio-mimetic simulation fidelity: **Cadaver-level mechanical realism** • AI procedural accuracy: **≥ 95% confidence compared to expert benchmarks** • Procedural success rate by non-experts: **≥ 90% within three attempts**			
R&D Plan	**Phase 1 (Year 1): Core Design and Material Optimization** • Design of AI-integrated training framework • Optimization of multi-branch, support-free vascular simulation materials **Phase 2 (Year 2): Prototype Development and Algorithm Refinement** • Development of high-precision integrated simulator prototypes • Refinement of AI guidance algorithms aligned with global standards **Phase 3 (Year 3): Validation and Standardization** • Empirical validation of sutureless rapid-connect modules • Establishment of international disaster response training protocols			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Development of AI-powered immersive trauma training modules • Empirical validation of procedural accuracy and time-to-hemostasis reduction • Alignment of training curricula with international emergency and trauma guidelines • System integration and pilot-level validation

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Expected R&R of Korean SME Partner	• Development of AI-powered immersive trauma training modules • Empirical validation of procedural accuracy and time-to-hemostasis reduction • Alignment of training curricula with international emergency and trauma guidelines • System integration and pilot-level validation

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected Outcomes** • Establishment of a new standard for AI-powered integrated trauma training • Validation of support-free, multi-branch vascular simulation technology • Demonstration of rapid, non-expert emergency vascular stabilization within the golden hour **Industrial, Economic, and Societal Impact** • Reduction of preventable mortality in disaster and mass-casualty environments • Strengthening of competitiveness in advanced bio-convergent medical training technologies • Expansion into global disaster relief and public healthcare procurement markets • Contribution to the formation of a high-value ecosystem combining AI, precision manufacturing, and emergency medicine

6. References

1. Emergency Vascular Intervention & Hemorrhage Control

- **TCCC Guidelines (2024):** "Preventable Death" in combat and disaster zones, emphasizing immediate junctional and extremity hemorrhage control.
- **Journal of Trauma and Acute Care Surgery:** Empirical studies on the "Golden Hour" in vascular injuries, highlighting the necessity of **sutureless rapid-connect technologies** for non-expert responders.
- **Advanced Bio-Materials:** Research on **support-free 3D bio-printed scaffolds** for immediate vascular replacement in emergency and disaster settings.

2. AI-Powered Medical Training & Simulation

- **IEEE Transactions on Medical Robotics and Biorobotics:** Analysis of AI-based real-time feedback systems for assessing the accuracy of surgical tasks by non-clinicians.
- **Nature Medicine:** The role of virtual/physical integrated simulation platforms in reducing procedural errors during high-stress disaster triage

3. Fluid Resuscitation & Infection Control

- **The Lancet – Emergency Medicine:** Comparative studies on pre-hospital fluid resuscitation and the impact of rapid IV/IO access on survival rates in mass casualty events.
- **World Journal of Emergency Surgery:** Protocols for preventing secondary infections and sepsis in contaminated disaster environments through rapid wound closure.